FEASIBILITY STUDY

FloodSense: An Expert System-Based Flood Susceptibility Android Application for Pre-Event Flood Preparedness in Bacoor City

Presented to the Department of Computer Studies

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OVERVIEW

Flooding remains one of the most persistent and consequential natural hazards affecting Bacoor City, Cavite. Located in a low-lying coastal area within Cavite province, Bacoor City is regularly exposed to flooding driven by monsoon rainfall, storm surges, and inadequate drainage infrastructure. As supported by the findings of the primary data gathering phase of this study, the impact of flooding on residents is not only physical but also informational — residents consistently lack timely, accurate, and location-specific flood susceptibility information that would allow them to make informed decisions before and during flood events.

A structured survey administered to 50 residents across 12 flood-prone barangays in Bacoor City revealed significant gaps in the current state of flood information dissemination. Of all respondents, 92% reported experiencing flooding in their area, with 54% experiencing it almost every rainy season and 58% reporting severe flooding. Despite the frequency and severity of flooding, only 20% of respondents considered themselves well-informed about flood-prone areas in Bacoor City — with the remaining 80% falling under 'Somewhat Aware,' 'Not Sure,' or 'Not Aware at All.' The three Likert-scale items on flood information quality (Q9–Q11) yielded mean scores of 2.62, 2.70, and 2.72 respectively, indicating near-dissatisfied perceptions of current information quality, timeliness, and clarity. Most critically, 64% of respondents reported having made a wrong decision due to insufficient flood information, including taking flooded roads, failing to evacuate in time, and misjudging flood conditions.

Thematic analysis of open-ended survey responses confirmed the nature of this gap. The two most prevalent themes from respondents regarding their biggest problem in getting flood information were Late or Delayed Information (26%) and Lack of Information or No Warnings at All (20%), together accounting for 38% of all responses. Respondents also cited unclear or inaccurate information (12%) and lack of location-specific flood susceptibility information (6%) as recurring barriers. These findings were validated through a Key Informant Interview with a staff member of the Bacoor Disaster Risk Reduction and Management Office (BDRRMO), who confirmed that the office's current flood information dissemination relies primarily on social media, email, and barangay coordination — and acknowledged that resident engagement with official warning channels remains low due to poverty, work constraints, and low attendance at flood orientations.

The BDRRMO interview further revealed that while the office maintains historical flood data and field-assessed exposure maps per barangay, these datasets are not accessible to the general public in a usable or interactive format. The interviewee confirmed that the office's data holdings include barangay boundary maps, historical flood records, and field-assessed susceptibility data — and expressed openness to data sharing for research and academic purposes. The BDRRMO officer affirmed that a Android-based tool providing location-specific flood susceptibility information and expert-guided preparedness advice would serve a genuine community need.

A supplementary problem-identification interview conducted with the Officer-in-Charge of the Bacoor City Planning and Development Office (CPDCO) confirmed that the office holds land-use maps, elevation maps, flood-hazard maps, and barangay boundary maps that can be released to the research team under a formal endorsement. The CPDCO officer further confirmed that Bacoor functions as a regional catch basin and that drainage degradation is a key flood driver, validating the environmental features selected for the expert knowledge base.

FloodSense is a proposed Android application that addresses these identified gaps through the integration of an Expert System-based flood susceptibility classification engine, a pre-rendered scenario-driven susceptibility map, and an active pre-event flood preparedness guidance system. Unlike the original design, which relied on machine learning for susceptibility classification and augmented reality for visualization, the revised FloodSense system is grounded entirely in expert knowledge formally sourced from BDRRMO — the institutional partner of this study. The thesis adviser confirmed that machine learning is more appropriate for learning behavioral patterns from data, while the susceptibility classifications required by FloodSense are factual, expert-validated assessments that are better encoded as structured expert system rules.

The Android application is built using Flutter (Dart), a purpose-built Android development framework that replaces Unity, which was originally selected for its augmented reality capabilities. With the AR component removed, Flutter offers a more efficient, maintainable, and Android-appropriate development environment for the revised feature set. Firebase continues to serve as the backend infrastructure for authentication, data storage, and serverless functions.

In response to the panel's challenge — 'What happens next after users become aware?' — three action features are incorporated: Check My Area, which presents an interactive map with a draggable pin for manual location identification without requiring GPS permissions; an Expert System DSS with an integrated AI assistant that presents a guided decision tree informed by BDRRMO expert knowledge and allows users to ask flood awareness questions in context; and Find Nearest Evacuation Center, which ranks BDRRMO-designated centers by distance and links to Google Maps. Together, these features transition FloodSense from a passive awareness tool to an active, expert-guided flood preparedness system.

KEY FEATURES

Core Features (Revised)

Pre-Rendered Flood Susceptibility Map

Displays a pre-rendered, color-coded barangay-level flood susceptibility map of Bacoor City. The map updates automatically based on the user's selected rainfall scenario (intensity and duration). No camera, augmented reality engine, or GPS anchoring is required. Users can pinch-to-zoom to examine individual barangay labels and susceptibility colors.

- Color-coded susceptibility levels: Low (green), Moderate (yellow), High (orange), Very High (red)
- Map updates dynamically based on the selected rainfall intensity and duration
- No camera or GPS required — full-city static visualization rendered in Flutter
- Color legend and disclaimer accessible at all times during map view
- Disclaimer: susceptibility levels represent expert-assessed estimates, not real-time flood guarantees

PAGASA-Aligned Rainfall Input

Users select a rainfall scenario and duration from dropdown menus using plain-language labels tied to PAGASA's public rainfall advisory classification. The app internally maps these selections to millimeter-per-hour values, removing the need for technical numeric input from non-expert users. This design was directly validated by the BDRRMO key informant.

- Rainfall intensity presets: Light (<2.5 mm/hr), Moderate (2.5–7.5 mm/hr), Heavy (7.5–15 mm/hr), Intense (15–30 mm/hr), Torrential (>30 mm/hr)
- Duration dropdown: 1 hr, 3 hrs, 6 hrs, 12 hrs, 24 hrs
- Saved rainfall preferences auto-filled from previous session

Expert System Flood Susceptibility Classification

Flood susceptibility per barangay is derived from expert knowledge formally sourced from BDRRMO, incorporating historical flood records, field-assessed exposure data, elevation features from CPDCO, and PAGASA rainfall parameters. Results are pre-computed and stored as static JSON lookup tables in Firebase Firestore. No machine learning model is used and no live inference occurs at runtime. The Expert System replaces the previously proposed ML classifier in accordance with the thesis adviser's guidance that factual susceptibility data is more appropriately encoded through expert knowledge rules than learned from behavioral patterns.

- Classification output: Low / Moderate / High / Very High — one per barangay per rainfall scenario
- Knowledge source: BDRRMO historical flood records, field-assessed barangay exposure data, CPDCO elevation and land-use maps, PAGASA rainfall classification tables
- Runtime behavior: app retrieves pre-computed JSON from Firestore — no ML server or model inference required

Results Summary Screen

A scrollable list of all Bacoor City barangays with color-coded susceptibility badges, populated from the pre-computed Firestore lookup table. Provides an at-a-glance overview without requiring the map interface.

User Account and Preference System

Users create an account to access personalized features including saved rainfall preferences. Firebase Authentication handles user management with secure credential storage.

- Username and password account creation and login
- Home barangay registered at account creation
- Saved rainfall preference parameters (intensity and duration)

Onboarding and Disclaimer Screen

A first-launch onboarding screen informs users of the app's purpose, limitations, and proper usage. Re-accessible from the main dashboard at any time.

- App purpose and scope explanation
- Disclaimer: 'This app provides expert-assessed flood susceptibility estimates. It is not a real-time flood sensor. Always follow official warnings from BDRRMO and PAGASA.'
- Introduction to the Expert System DSS and AI assistant features

Action Features (Post-Panel Feedback)

In response to the panel's challenge — 'What happens next after users become aware?' — three action features transition FloodSense from a passive awareness tool to an active, expert-guided flood preparedness system. All three operate within the existing Firebase free tier or minimal-cost API tiers.

Check My Area — Location-Based Safety Status

After viewing the susceptibility map or Results Summary, the user taps 'Check My Area'. The app presents an interactive map of Bacoor City with a draggable pin marker. The user manually drags and drops the pin onto their preferred location — no GPS permission is requested or required. The app cross-references the pin's coordinates against a static GeoJSON barangay boundary file to identify the user's barangay, then queries the pre-computed Firestore susceptibility lookup for that barangay under the currently selected rainfall scenario. A dedicated results screen displays the barangay name, susceptibility level with color indicator, active rainfall scenario, and a plain-language safety status summary.

- All pin coordinate processing performed locally on-device — not stored or transmitted
- Compliant with Republic Act 10173 (Data Privacy Act)
- Result is scenario-aware: changing the rainfall scenario refreshes the result for the same pinned location automatically

Expert System DSS — Decision Tree with Integrated AI Assistant

Automatically displayed after the Safety Status card. The Expert System presents a structured decision tree that guides the user through a series of situational questions and branches to level-appropriate, BDRRMO-informed action recommendations. Integrated within the same interface is an AI assistant trained on BDRRMO expert knowledge and pre-event flood awareness data. At every step of the decision tree, users may ask open-ended flood awareness questions and receive expert-grounded, contextually relevant responses informed by their current pinned barangay, susceptibility level, and active rainfall scenario.

- Decision tree knowledge base sourced exclusively from BDRRMO expert data — not general internet sources

- AI assistant is context-aware: receives user's barangay, susceptibility level, and rainfall scenario with each query
- Example queries: 'What should I put in a go-bag?', 'Which roads in Talaba flood during heavy rain?', 'What does Very High susceptibility mean for my family?'
- Decision tree functions fully offline using locally stored rules; AI assistant requires internet connection and falls back gracefully when unavailable
- Compliant with RA 10173 — no personally identifiable information transmitted with AI queries

Find Nearest Evacuation Center

Triggered automatically when the user's barangay is classified as Moderate, High, or Very High susceptibility. A ranked list of BDRRMO-designated evacuation centers is displayed, sorted by straight-line distance from the user's pinned location using the Haversine formula computed on-device. Each entry shows the center name, barangay, and estimated distance. The user taps 'Get Directions' to be deep-linked into Google Maps for turn-by-turn navigation.

- Evacuation center list sourced from BDRRMO as part of the existing data-sharing collaboration
- No routing API or paid geolocation service required
- All navigation handled by Google Maps via standard geo: URI deep-link
- Distance calculated from the pin coordinates set in Check My Area — no additional location input required

I. TECHNICAL FEASIBILITY

This section evaluates whether FloodSense can be built using currently available technologies, tools, and frameworks within the capability of an undergraduate Computer Science team. Each component is mapped to a specific, free or low-cost technology with a documented implementation path. The assessment covers the revised core features — including the transition from Unity and Vuforia to Flutter, the removal of the machine learning pipeline, and the introduction of the Expert System DSS with an integrated AI assistant.

1.1 Proposed Technology Stack

Layer	Technology	Cost & Justification
Android App (All Screens)	Flutter (Dart)	Free (open-source). Purpose-built for multi-screen Android applications. Replaces Unity + uGUI, which was selected solely for AR capabilities no longer required in the revised system. Flutter's widget system, hot-reload, and Firebase SDK integration make it more efficient for the revised FloodSense architecture.
Map Display	flutter_map (open-source) or Google Maps Flutter plugin	Free. Supports pre-rendered color-coded barangay overlays, GeoJSON boundary rendering, and draggable pin markers. Replaces Vuforia ground-plane AR tracking.
Backend / Database	Firebase (Firestore, Auth, Functions)	Free tier (Spark Plan). Real-time sync, built-in authentication, cloud storage, and serverless functions. Unchanged from original design.
Expert System Knowledge Base	Firebase Firestore (pre-computed JSON lookup tables)	Free tier. BDRRMO expert knowledge encoded as static susceptibility lookup tables keyed by rainfall intensity and duration. Replaces ML model and scikit-learn pipeline — no training infrastructure required.
DSS Decision Tree	Rule-based logic (Dart) + Firestore	Free. Decision tree rules encoded from BDRRMO expert knowledge and stored in Firestore. Rendered in Flutter widgets. No external API required for the decision tree layer.
AI Assistant	LLM API (e.g., Anthropic Claude API) + Firebase Functions	Pay-per-use (minimal cost for pilot scale of 20–50 users). LLM trained via system prompt embedding BDRRMO expert knowledge. Firebase Functions serve as secure middleware to handle API calls without exposing credentials in the app.
GeoJSON Boundary Lookup	Static GeoJSON file (bundled with app)	Free. Used to resolve pin coordinates to barangay names. Processed on-device — no external API call required.

Distance Calculation	Haversine formula (Dart, on-device)	Free. Calculates straight-line distance from pin coordinates to each evacuation center. No routing API required.
Navigation (Directions)	Google Maps URI deep-link	Free. Passes destination coordinates to the installed Google Maps app. All routing handled by Google Maps.
Elevation / Spatial Data	CPDCO (via data-sharing agreement)	Free (institutional). Topographic and land-use features for expert knowledge encoding.
Historical Flood Data	BDRRMO (via data-sharing agreement)	Free (institutional). Ground-truth flood occurrence data per barangay; evacuation center list with GPS coordinates.
Rainfall Reference	PAGASA Advisory Classification	Free (public domain). Basis for user-facing rainfall intensity presets.
UI/UX Prototyping	Figma	Free tier. Android interface design and prototyping prior to Flutter development.

1.2 Feature-Level Technical Assessment

Feature	Implementation Approach	Complexity	Est. Time
Pre-Rendered Susceptibility Map	flutter_map or Google Maps Flutter plugin; color-coded barangay zone overlay from pre-computed Firestore lookup; pinch-to-zoom interaction	Low	1 week
PAGASA Rainfall Input + Firestore Lookup	Flutter Dropdown widgets mapped to mm/hr values; app reads matching pre-computed susceptibility	Low	1 week

	JSON from Firestore		
Expert System Classification (Firestore)	BDRRMO expert knowledge encoded as pre-computed JSON lookup tables; uploaded to Firestore; Flutter app fetches the appropriate document. No ML server required.	Low	1 week
Results Summary Screen	Flutter ListView of barangay names with color badge; fed from ML lookup stored in Firestore	Low	0.5 week
User Account + Auth (Firebase)	Firebase Authentication for username/password login; Firestore for user preferences and saved barangay	Low	1 week
Onboarding + Disclaimer Screen	Flutter first-launch screen with instructions and disclaimer; dismissable; re-accessible from main dashboard	Low	0.5 week
Check My Area	flutter_map or Google Maps Flutter plugin with draggable marker; GeoJSON	Low-Medium	1 week

	boundary lookup using pin coordinates; Firestore susceptibility lookup; scenario-aware results screen. No GPS permission required.		
DSS Decision Tree	Rule-based decision tree encoded from BDRRMO knowledge; Flutter widget-based step-by-step question interface; Firestore for knowledge base storage	Medium	1.5 weeks
AI Assistant (integrated in DSS)	LLM API with BDRRMO-grounded system prompt; Firebase Functions middleware; Flutter chat widget embedded in the DSS interface; context injection per query (barangay, susceptibility level, rainfall scenario)	Medium	1.5 weeks
Find Nearest Evacuation Center	Static evacuation center JSON from BDRRMO in Firestore;	Low-Medium	1 week

	Haversine formula for on-device distance calculation; ranked Flutter ListView; 'Get Directions' deep-links to Google Maps via geo: URI		

1.3 Technical Challenges and Mitigations

Flutter Learning Curve

The team's prior experience was with Unity (C#). Transitioning to Flutter (Dart) introduces a new language and framework.

Mitigation: Dart is syntactically similar to Java and C#, reducing the learning curve for a CS team. Flutter's extensive documentation, large developer community, and hot-reload feature significantly shorten iteration time. The team is advised to complete Flutter fundamentals during Phase 1 of the development timeline.

Expert System Knowledge Encoding

Translating BDRRMO expert knowledge into structured susceptibility lookup tables and decision tree rules requires careful data collection, validation, and encoding.

Mitigation: Conduct structured knowledge elicitation sessions with BDRRMO using predefined templates aligned with the lookup table schema (barangay × rainfall scenario × susceptibility level). Cross-validate encoded rules against BDRRMO's existing field-assessed exposure data and historical flood records. All encoded knowledge must be reviewed and approved by BDRRMO before deployment.

AI Assistant Knowledge Grounding

An LLM API trained only on its general training data may produce responses that are inaccurate, generic, or irrelevant to Bacoor City's specific flood conditions.

Mitigation: The AI assistant is grounded through a carefully curated system prompt that embeds BDRRMO expert knowledge — including barangay-level flood history, preparedness protocols, evacuation procedures, and locally relevant road and area information. The system prompt is treated as the AI's authoritative knowledge base. Responses are constrained to flood awareness and preparedness topics relevant to Bacoor City. The AI assistant is tested against a set of expected queries before deployment to evaluate response accuracy and relevance.

AI Assistant Internet Dependency

The AI assistant requires an active internet connection for each query. During flood events, connectivity may be degraded or unavailable.

Mitigation: The decision tree layer of the Expert System DSS operates fully offline using locally stored rules in Firestore's offline persistence cache. When the AI assistant is unavailable due to connectivity issues, a graceful fallback message is displayed directing users to the decision tree for guidance. The core susceptibility lookup and Check My Area features are also fully functional offline.

Pin Placement Accuracy

Users placing the draggable pin near barangay boundary edges may occasionally resolve to an adjacent barangay due to minor GeoJSON digitization inaccuracies.

Mitigation: Cross-validate the GeoJSON boundary file against NAMRIA and CPDCO reference data. Inform users via the onboarding screen that boundary detection is approximate and that they should confirm their barangay manually if the detected result appears incorrect. The draggable pin design gives users direct visual control over their queried location, reducing ambiguous boundary placements compared to automatic GPS detection.

GeoJSON Boundary Accuracy

Barangay boundary GeoJSON files derived from hard-copy maps may contain minor digitization inaccuracies.

Mitigation: Cross-validate the GeoJSON boundary file against NAMRIA and CPDCO reference data. Inform users via the onboarding screen that boundary detection is approximate.

The draggable pin interaction allows users to make manual corrections if the detected barangay appears incorrect.

II. OPERATIONAL FEASIBILITY

This section evaluates whether FloodSense can be realistically developed, deployed, and used within the constraints of the thesis timeline, team size, target users, and available institutional resources. The assessment reflects the fully revised feature set, including the transition to Flutter, the Expert System approach, and the integrated AI assistant.

2.1 Team Composition and Role Distribution

Member	Primary Role & Responsibilities
Member 1 — Android Developer	Flutter app development: all screens (login, dashboard, map display, results summary, check my area, DSS decision tree, AI assistant chat, evacuation center, account); Flutter map widget integration; draggable pin implementation; GeoJSON boundary lookup; Firebase integration on the client side.
Member 2 — Backend / Expert System Engineer	Expert system knowledge encoding: structuring BDRRMO data into susceptibility lookup tables and decision tree rules; Firebase setup; Firestore data upload and management; Firebase Functions development for AI assistant middleware; LLM API integration and system prompt engineering.
Member 3 — Data / Research Lead	Spatial data collection and preprocessing (CPDCO, BDRRMO); barangay zone GeoJSON preparation; evacuation center data encoding; knowledge elicitation sessions with BDRRMO; usability testing coordination; thesis documentation.

2.2 Recommended Development Timeline (10 Months)

Phase / Period	Deliverables
Phase 1 — Foundation (Months 1–2)	Project setup; Firebase configuration; user authentication; Flutter framework onboarding; basic Flutter UI screens; formal data

	request letters to BDRRMO and CPDCO; initial data preprocessing; GeoJSON barangay boundary file acquisition or preparation.
Phase 2 — Expert System Development (Months 2–4)	Structured knowledge elicitation sessions with BDRRMO; encoding of susceptibility lookup tables (barangay × rainfall scenario × susceptibility level) into Firestore JSON; encoding of decision tree rules from BDRRMO preparedness protocols; AI assistant system prompt development and testing using BDRRMO expert knowledge; Firebase Functions middleware for LLM API; encoding of BDRRMO evacuation center list into Firestore.
Phase 3 — Map & Core Features (Months 4–6)	Pre-rendered barangay susceptibility map integration using flutter_map or Google Maps Flutter plugin; color-coded zone overlay; scenario-driven map update logic; results summary screen; PAGASA-aligned rainfall input; user preferences and account management.
Phase 4 — Action Features Integration (Months 6–8)	Check My Area: flutter_map draggable pin, GeoJSON boundary lookup, scenario-aware results screen; DSS Decision Tree: Flutter step-by-step question interface with Firestore-backed rules; AI Assistant Chat: Flutter chat widget integrated into DSS interface, Firebase Functions middleware, context injection; Find Nearest Evacuation Center: Haversine distance calculation, ranked list UI, Google Maps deep-link.
Phase 5 — Testing & Refinement (Months 8–9)	Usability testing with at least 20 residents from the survey pool; BDRRMO review of susceptibility outputs, decision tree rules, AI assistant response quality, and evacuation center data accuracy; bug fixes; final feature refinements; AI assistant response evaluation against expected query set.
Phase 6 — Final Defense Preparation (Month 10)	Final defense preparation; documentation completion; thesis manuscript finalization.

2.3 Target Users and Adoption Feasibility

FloodSense targets all residents of Bacoor City, with priority distribution through BDRRMO community outreach channels and direct recruitment from the original survey respondent pool.

- Tier 1 — Bacoor City Residents (flood-prone barangays): Residents of historically susceptible barangays (Zapote, Mambog, Talaba, Niog) who experience flooding regularly and need location-specific susceptibility information and expert-guided preparedness advice for evacuation and safety decisions. Pilot distribution through BDRRMO community outreach and barangay-level orientation; direct recruitment from survey respondents.
- Tier 2 — General Bacoor Residents: Residents outside the highest-susceptibility zones who use the app for general flood awareness and community preparedness planning. App store distribution; awareness through city-level flood information campaigns.

2.4 Financial Feasibility

FloodSense is designed to operate at near-zero licensing cost during the thesis development and pilot testing phases. The transition to Flutter and the Expert System approach eliminates the costs associated with Unity Pro licensing considerations and ML cloud inference. The only service with a usage-based cost is the LLM API for the AI assistant, which is estimated to be minimal at pilot scale.

Tool / Service	Purpose	Cost
Flutter SDK	Android app development framework for all screens and UI	Free (open-source)
flutter_map / Google Maps Flutter Plugin	Pre-rendered susceptibility map overlay and draggable pin interaction	Free
Firebase Spark Plan	Authentication, Firestore database, Firebase Functions, offline persistence	Free tier
LLM API (e.g., Anthropic Claude API)	AI assistant powering the Expert System DSS conversational interface, grounded on BDRRMO expert knowledge	Pay-per-use; estimated minimal cost at pilot scale (20–50 users)

Google Maps URI Deep-Link	Turn-by-turn navigation from Find Nearest Evacuation Center	Free
CPDCO Topographic Data	Elevation and land-use features for expert knowledge encoding	Free (data-sharing agreement)
BDRRMO Historical Flood Data	Ground-truth barangay flood records; evacuation center list; expert knowledge for DSS and AI assistant	Free (data-sharing agreement)
PAGASA Rainfall Advisory Data	Rainfall intensity classification reference for dropdown input	Free (public domain)
Figma	UI/UX prototyping prior to Flutter development	Free tier

2.5 Operational Risks and Mitigations

BDRRMO Data Access Delay

BDRRMO may require additional time to process a formal data-sharing request, potentially delaying expert system knowledge encoding and evacuation center data entry.

Mitigation: Initiate formal data request immediately after title defense approval. Begin expert system development in parallel using publicly available PAGASA data and CPDCO maps as primary sources. Placeholder susceptibility values and decision tree rules can be used for development purposes, with official BDRRMO data substituted upon release.

CPDCO Spatial Data Inaccessibility

CPDCO may not have digital GIS files readily available, or may store data in formats incompatible with the research pipeline.

Mitigation: Hard-copy maps can be georeferenced and digitized using QGIS. If digitization time exceeds the project schedule, open digital elevation models (SRTM, ALOS PALSAR) and NAMRIA land-cover data will be used directly.

AI Assistant Response Quality

The AI assistant may produce responses that are too generic, inaccurate for Bacoor City's specific conditions, or outside the intended flood awareness scope.

Mitigation: The AI assistant's system prompt is treated as its authoritative knowledge base and is developed iteratively using structured BDRRMO expert knowledge. A dedicated evaluation phase (Phase 5) tests the assistant against a set of expected user queries covering all susceptibility levels, barangays, and common flood awareness topics. BDRRMO staff review AI responses for factual accuracy before deployment. Response scope is restricted to flood awareness and preparedness topics relevant to Bacoor City.

Low Pilot User Adoption

Community residents, particularly elderly or low-literacy users, may find the app interface or the AI assistant chat unfamiliar or difficult to use.

Mitigation: Conduct usability testing with at least 20 residents from the original survey pool. Simplify onboarding with visual step-by-step instructions. The DSS decision tree uses simple tap-to-answer interactions before offering the AI chat, reducing the initial interaction complexity. The draggable pin eliminates the need for GPS permissions, removing a common Android app friction point. Plain-language labels (aligned with PAGASA classifications) are used throughout.

Scope Management

The integration of Flutter, an Expert System, a DSS decision tree, and an AI assistant within a single thesis represents a substantial scope.

Mitigation: The phased timeline isolates Flutter onboarding (Phase 1), expert system encoding (Phase 2), map and core feature development (Phase 3), and action feature integration (Phase 4) into distinct, manageable phases. The removal of the ML training pipeline and AR integration from the original design significantly reduces technical complexity. The decision tree and AI assistant share the same Firebase Functions infrastructure, reducing duplication of effort.

Data Privacy Compliance

User account data and AI assistant queries are subject to Philippine Republic Act 10173 (Data Privacy Act).

Mitigation: Firebase Security Rules enforce per-user data isolation. Informed consent obtained from all pilot participants. Pin coordinates used in Check My Area and Find Nearest Evacuation Center are processed on-device only and not stored or transmitted. AI assistant queries are routed through Firebase Functions middleware and contain no personally identifiable information — only the contextual scenario data (barangay name, susceptibility level, rainfall scenario). The draggable pin design eliminates the need to request GPS permissions entirely.

Sustainability Post-Thesis

Without a formal maintenance agreement, the app's expert knowledge base may become outdated as new flood events occur and evacuation center data changes.

Mitigation: Frame sustainability as future work in the thesis defense. Proposed sustainability pathway: establish a formal MOU with BDRRMO for periodic knowledge base updates using updated historical flood records and revised preparedness protocols. Explore integration with BDRRMO's existing community information dissemination channels for sustained distribution. LLM API costs for full public deployment to be evaluated and proposed to BDRRMO or the university for institutional support.

Interview-Identified Gap: User Input Accuracy

As raised by the BDRRMO key informant, users cannot accurately estimate rainfall in millimeters, making free-text or numeric input unreliable.

Mitigation: Fully addressed through the PAGASA-aligned dropdown design — users select plain-language intensity levels (Light, Moderate, Heavy, Intense, and Torrential) tied to standardized mm/hr ranges. The app handles all technical conversion internally. This design decision was directly validated by the BDRRMO interview.

Interview-Identified Gap: Output Clarity for Non-Expert Users

The BDRRMO interviewee raised that users unfamiliar with susceptibility mapping may misread susceptibility levels as absolute flood predictions rather than expert-assessed estimates.

Mitigation: Include a brief legend explanation accessible from the map view. Add disclaimer on first launch: 'Susceptibility levels indicate the likelihood of flooding based on expert assessment of environmental conditions — not a guarantee of actual flooding.' The plain-language safety status cards in Check My Area reinforce this framing. The AI assistant is also instructed to clarify the meaning of susceptibility levels when asked.

III. FEASIBILITY SUMMARY

Dimension	Assessment	Verdict
Technical	All features implementable with free, well-documented tools. Expert System susceptibility classification uses BDRRMO-sourced pre-computed lookup tables stored in Firestore — no ML server required. Flutter replaces Unity as the Android framework, better suited for a multi-screen, data-driven app. Pre-rendered map replaces AR (Vuforia removed). DSS decision tree uses rule-based Dart logic with Firestore. AI assistant uses LLM API with Firebase Functions middleware. GeoJSON boundary lookup and Haversine distance calculation performed on-device.	FEASIBLE
Financial	Near-zero licensing cost during thesis period. Flutter, Firebase Spark Plan, flutter_map, and Google Maps URI deep-link are all free. LLM API (AI assistant) operates on a pay-per-use model estimated to be minimal at pilot scale of 20–50 users. Removal of Unity, Vuforia, and ML cloud infrastructure eliminates previously anticipated licensing and compute costs.	FEASIBLE
Team Capacity	Three-person team with clearly separated Android Dev, Backend/Expert System, and Data/Research roles. 10-month phased timeline isolates Flutter	FEASIBLE

	onboarding, expert system encoding, map and core feature development, and action feature integration into distinct phases to prevent bottlenecks. Flutter learning curve is manageable given the team's C#/Java background.	
User Adoption	Pilot testing achievable through BDRRMO community outreach and direct recruitment from survey respondents. PAGASA-aligned plain-language input removes technical barrier. Decision tree presents structured guidance before the AI chat, minimizing complexity for non-technical users. Draggable pin eliminates GPS permission friction. AI assistant grounded on BDRRMO knowledge ensures locally relevant responses.	FEASIBLE
Institutional Support	BDRRMO key informant confirmed openness to data sharing and collaboration; serves as the institutional expert knowledge source for both the Expert System classification and the AI assistant. CPDCO confirmed required spatial maps exist and can be released as hard-copy prints. BDRRMO evacuation center data obtainable under the existing data-sharing collaboration.	FEASIBLE
Scope	Revised system removes ML training pipeline and AR engine (Vuforia, Unity) — significant reduction in technical complexity vs. original design. New additions (Expert System DSS with integrated AI assistant) are architecturally grounded in a single Firebase backend and share the same Firestore data backbone. Combined estimated development time for all new and revised features is approximately 7–8 weeks, well within the 10-month window.	FEASIBLE
Academic Contribution	Novel integration of Expert System-based flood susceptibility classification, pre-event decision support (DSS), and an AI assistant grounded in institutional expert knowledge for a Philippine LGU context. PAGASA-aligned community input design. BDRRMO-validated problem, concept, and knowledge base. Extends the study beyond susceptibility mapping	FEASIBLE

	into actionable, expert-guided community-level flood preparedness.	